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# Introduction

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Semester → III
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Assignment No → 1
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# XOR Alarm Problem (Perceptron)

Given: $z = w_1x_1 + w_2x_2 + b$ and $H(z) = 1$ if $z \geq 0$, 0 if $z < 0$

Alarm = 1 only when exactly one sensor is ON

So it is XOR.

Decision boundary of a Perceptron: $w_1x_1 + w_2x_2 + b = 0$ (a straight line).

## (i) four inequalities.

Rule: if $y = 1$ we need $z \geq 0$, if $y = 0$ we need $z < 0$.

| $x_1$ | $x_2$ | $y$ | $z$ | Condition |
|---|---|---|---|---|
| 0 | 0 | 0 | $b$ | $b < 0$ ... (i) |
| 0 | 1 | 1 | $w_2 + b$ | $w_2 + b \geq 0$ ... (ii) |
| 1 | 0 | 1 | $w_1 + b$ | $w_1 + b \geq 0$ ... (iii) |
| 1 | 1 | 0 | $w_1 + w_2 + b$ | $w_1 + w_2 + b < 0$ ... (iv) |

## (ii) Proof: no $w_1, w_2, b$ works

Proof A (algebra). Assume (i) to (iv) all hold.

Add (ii) + (iii): $w_1 + w_2 + 2b \geq 0$ ... (v)

from (iv): $w_1 + w_2 < -b$. Put in (v):

$$w_1 + w_2 + 2b < -b + 2b = b$$

By (i) $b < 0$, therefore $w_1 + w_2 + 2b < b < 0$.

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But (v) says it is $\geq 0$, contradiction.

Proof B (geometry, midpoint). $z$ is a linear function
So
$z(\frac{1}{2}P + \frac{1}{2}Q) = \frac{1}{2}z(P) + \frac{1}{2}z(Q)$
Midpoint of $(0,1)$ and $(1,0)$ is $M = (\frac{1}{2}, \frac{1}{2})$:
$z(M) = \frac{1}{2}(w_2+b) + \frac{1}{2}(w_1+b) = (w_1+w_2+2b)/2 \geq 0$

Midpoint of $(0,0)$ and $(1,1)$ is the same $M$
$z(M) = \frac{1}{2}(b) + \frac{1}{2}(w_1+w_2+b) = (w_1+w_2+2b)/2 < 0$
Same point $M$ cannot give $z \geq 0$ and $z < 0$.
So the two classes are not linearly separable (their convex hulls cross at $M$).

Best a single perceptron can do is 3 out of 4 (75%.)
Example: $w_1 = w_2 = 1, b = -\frac{1}{2}$ gives outputs 0, 1, 1, 1
last row wrong.

(iii) Network: 2 hidden + 1 output neuron

Idea: $XOR = (x_1 \text{ OR } x_2) \text{ AND NOT } (x_1 \text{ AND } x_2)$

Hidden $h_1$ = OR. Need: $b_1 < 0$, $w+b_1 \geq 0$ (for 01 and 10), $2w+b_1 \geq 0$.
Take $w = 1, b_1 = -1$: $-1 < 0$ ✓, $0 \geq 0$ ✓, $1 \geq 0$ ✓

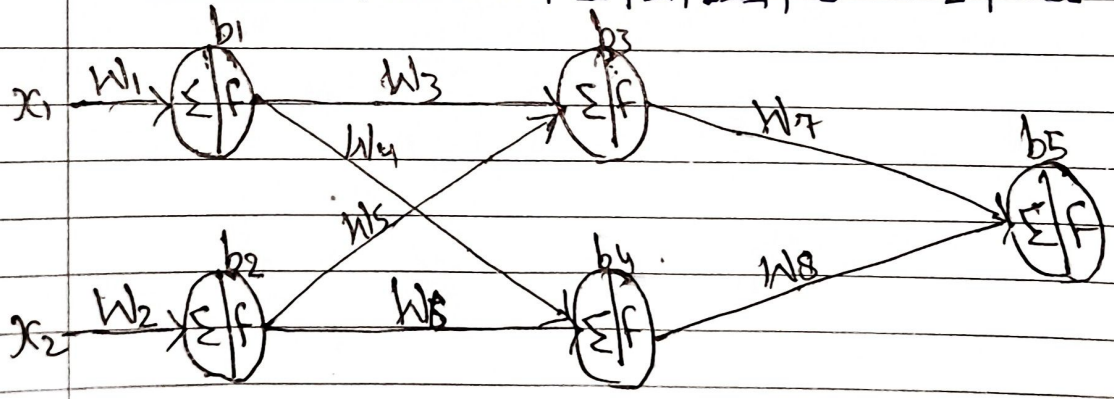
Hidden $h_2$ = AND. Need: $b_2 < 0$, $w+b_2 < 0$, $2w+b_2 \geq 0$
Take $w = 1, b_2 = -2$: $-2 < 0$ ✓, $-1 < 0$ ✓, $0 \geq 0$ ✓

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Output $y$: $(h_1, h_2)$ can be $(0,0), (1,0), (1,1)$ only
Need:
$(0,0) \to 0$: $b_3 < 0$; $(1,0) \to 1$: $v_1 + b_3 \geq 0$; $(1,1) \to 0$: $v_1 + v_2 + b_3 < 0$
Take $v_1 = 1, v_2 = -1, b_3 = -1$: $-1 < 0$ ✓; $0 \geq 0$ ✓; $-1 < 0$ ✓

| Neuron | Role | Weight | Bias |
|---|---|---|---|
| $h_1$ | OR | $w_{11}=1, w_{12}=1$ | $b_1 = -1$ |
| $h_2$ | AND | $w_{21}=1, w_{22}=1$ | $b_2 = -2$ |
| $y$ | $h_1$ AND NOT $h_2$ | $v_1 = 1, v_2 = -1$ | $b_3 = -1$ |

Matrix form: $h = H(Wx + b)$ $y = H(v^T h + b_3)$
$W = [[1,1],[1,1]]$, $b = [-1, -2]^T$, $v = [1, -1]^T$

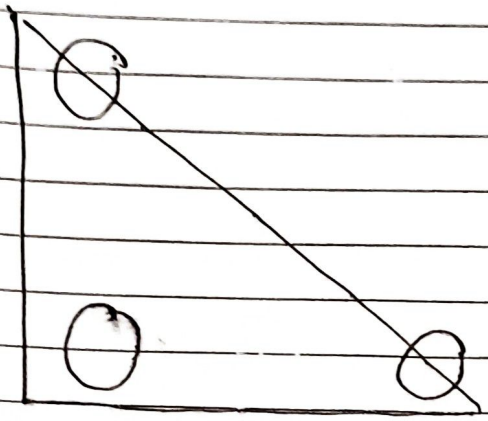



Why it works: hidden layer maps the points to a new space.

| $(x_1, x_2)$ | $(0,0)$ | $(0,1)$ | $(1,0)$ | $(1,1)$ |
|---|---|---|---|---|
| $(h_1, h_2)$ | $(0,0)$ | $(1,0)$ | $(1,0)$ | $(1,1)$ |
| $y$ | 0 | 1 | 1 | 0 |

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In $(h_1, h_2)$ Space the line $h_1 - h_2 = 1$ separates the classes, so it is linearly seperable here.



filled = alarm 1, empty = alarm 0, Alarm is 1 in the strip between the two lines $= 1 \leq x_1 + x_2 < 2$.

## (V) Forward Pass

$z_1 = x_1 + x_2 - 1, \; h_1 = H(z_1)$
$z_2 = x_1 + x_2 - 2, \; h_2 = (H(z_2)$
$z_3 = 1 \cdot h_1 - 1 \cdot h_2 - 1, \; y = H(z_3)$

| $x_1$ | $x_2$ | $z_1$ | $h_1$ | $z_2$ | $h_2$ | $z_3$ | $\hat{y}$ | $y$ | ✓ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | -1 | 0 | -2 | 0 | -1 | 0 | 0 | ✓ |
| 0 | 1 | 0 | 1 | -1 | 0 | 0 | 1 | 1 | ✓ |
| 1 | 0 | 0 | 1 | -1 | 0 | 0 | 1 | 1 | ✓ |
| 1 | 1 | 1 | 1 | 0 | 1 | -1 | 0 | 0 | ✓ |

Matrix check for $(1,1)$: $Wx + b = [2,2] + [-1,-2] = [1,0] \rightarrow h = [1,1]$

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$z_3 = 1(1) + (-1)(1) - 1 = -1 \rightarrow y = 0$

Matrix check for $(0,1)$: $wx + b = [1,1] + [-1,-2] = [0,-1] \rightarrow h = [1,0]$.

$z_3 = 1(1) + (-1)(0) - 1 = 0 \rightarrow y = H(0) = 1$

All four outputs match the truth table.

(V) Why 100,000 epochs will not help

Representability: does any weight set exist in this architecture that gives the required function?

Trainability: can gradient-based learning actually reach it.

(a) Representability failure → By Part (2), no $(w_1, w_2, b)$ gives 4/4 Perceptron rule: $w \leftarrow w + \eta(y - \hat{y})x$, $b \leftarrow b + \eta(y - \hat{y})$. Run with $\eta = 1$! $w = (0,0)$, $b = 0$.

| Epoch | Mistakes | w after | b after |
|---|---|---|---|
| 1 | 3 | (-1, 0) | -1 |
| 2 | 3 | (-1, 0) | 0 |
| 3 | 4 | (-1, 0) | 0 |
| 4, 5, ... 100000 | 4 | (-1, 0) | 0 |

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The weights fall into a loop and Mistakes never reach 0.
More epochs only repeat the same search inside the same set of lines. Convergence theorem needs linearly separable data.

(b) Trainability is a different issue

Even when a solution exists (our 2-hidden net), gradient descent must find it. Loss $L(\hat{y}, y)$ · chain rule:

for step function $H(z) = 0$ for $z \neq 0$ (undefined at 0.) · So $\partial L / \partial w = 0$.